

## Project Initialization and Planning Phase

|                      |                                     |
|----------------------|-------------------------------------|
| <b>Date</b>          | 01 Oct 2026                         |
| <b>Team ID</b>       | SWTID-2026-7456                     |
| <b>Project Name</b>  | ComicCraft - AI Comic Story Creator |
| <b>Maximum Marks</b> | 5 Marks                             |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform story creation using generative AI, letting users go from a simple idea to a fully illustrated comic. It tackles the time, skill and cost barriers of comic making. Key features include an AI story generator, automatic panel illustration and PDF export.

| Project Overview  |                                                                                                                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | To let anyone turn a short idea into a complete, illustrated comic story using generative AI, without needing writing or drawing skills, in a few minutes.                                                                                                                                                                                    |
| Scope             | A web application that takes a story idea, genre and panel count; generates an outline (Gemini Flash), a detailed story with panel descriptions (Gemini Pro) and one image per panel (Stable Diffusion); and exports the finished comic as a PDF.                                                                                             |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                               |
| Description       | Making a comic needs scripting, illustration and layout skills and usually takes days of effort. Students, teachers, writers and hobbyists often have story ideas but cannot turn them into visual stories quickly or affordably.                                                                                                             |
| Impact            | Solving this makes visual storytelling accessible to everyone, cuts creation time from days to minutes, lowers cost, and supports education, content creation and creative expression.                                                                                                                                                        |
| Proposed Solution |                                                                                                                                                                                                                                                                                                                                               |
| Approach          | A multi-stage generative AI pipeline: a fast model (Gemini Flash) drafts the outline, a stronger model (Gemini Pro) expands it into scenes, dialogue and image prompts, Stable Diffusion renders each panel, and FPDF assembles everything into a downloadable comic. FastAPI serves the workflow through a simple Jinja2 web interface.      |
| Key Features      | <ul style="list-style-type: none"> <li>- AI-generated story outline from a single prompt (Gemini Flash).</li> <li>- Detailed story and panel-by-panel scripts (Gemini Pro).</li> <li>- Automatic panel image generation (Stable Diffusion via Diffusers).</li> <li>- Comic preview in the browser and one-click PDF export (FPDF).</li> </ul> |

## Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                                                                                                      |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                                         |                                                                                                                               |
| Computing Resources     | CPU/GPU specifications, number of cores | T4GPU(16GBVRAM)forStableDiffusion; multi-core CPU                                                                             |
| Memory                  | RAM specifications                      | 16 GB                                                                                                                         |
| Storage                 | Disk space for data, models, and logs   | 50GBSSD(model weights, generated images, PDFs)                                                                                |
| <b>Software</b>         |                                         |                                                                                                                               |
| Frameworks              | Python frameworks                       | FastAPI, Uvicorn, Jinja2                                                                                                      |
| Libraries               | Additional libraries                    | google-generativeai, diffusers, torch, transformers, fpdf, Pillow, python-dotenv                                              |
| Development Environment | IDE                                     | VS Code, Jupyter Notebook / Google Colab                                                                                      |
| <b>Data</b>             |                                         |                                                                                                                               |
| Data                    | Source, size, format                    | No training dataset; pre-trained Gemini(API) and StableDiffusion(HuggingFace) models. Inputs: text prompts. Outputs: PNG, PDF |